Which language, and what that admits

A signal is read; a convention is declared; nothing is inferred. The whole design of this layer is that distinction.

1 · resolving the language

strongest signal first, stop at the first hit

1. declared

`language on /v1/search`

A deliberate statement about a corpus, made at READ time — so a vault ingested under one language answers correctly under another with no re-ingest.

Outranks everything below.

2. else the script

where a script settles it

Greek · Georgian · Hangul — one script, one language. A fact, not a reading.

Cyrillic → Russian, Devanagari → Hindi: the majority language, and labelled approximate because that is what it is.

3. else the drawer itself

`language_of_drawer`

A text carrying der, die, und, nicht is German. Only CLOSED-class words vote — a loanword does not get one.

Decisive or nothing: three hits and twice the runner-up, else stay silent.

Read per candidate, not per query: a vault may hold several languages, and the drawer is the unit that has one.

2 · five rules, every one of them pairwise

each answers about two strings and builds no class

`suffix_family`
SHAPE, not length. -er is German-only: Kind/Kinder, Haus/Häuser

`inflection_family`
A CLOSED set of endings per language, stem ≥ 3: libro/libri, книга/книги

`agglutinative_family`
Endings that stack: Turkish, Korean, Georgian, Hungarian

`ar_root_family`
144 roots × 20 patterns = 2859 forms. Two words meet only if ONE root fits.

`IRREGULAR`
~110 forms no rule over shape can reach: go/went, Mann/Männer

Why not a stemmer

A stemmer builds an equivalence class, and one false friend poisons every member. Measured: Snowball Greek merges πολύ (much) with πόλη (city). A pairwise rule costs only its own pair.

Why Arabic needed its own table

An ALLOWLIST: a form the table cannot generate matches nothing. Subsequence rules failed three times — بيت→بيوت and يجب→يجيب are the same string operation, and no rule over shape separates them.

3 · three channels, and only two of them are a claim about words

`lexical_exact`
The drawer said the word.
admits

`lexical_morph`
It holds a word BUILT ON it.
admits — and says so separately

`semantic`
Cosine, above the gate the EMBEDDER itself calibrates.

Measured — 191 paradigm pairs, 19 languages

lexical recall, declared	55.0% → 100%
lexical recall, undeclared	→ 100%
pairs resting on the embedder alone	20 → 0
negative controls	0 → 48

The prices, pinned rather than hidden

Declaring German is what makes -er legal, and -er also relates flow/flower, corn/corner, butt/butter. That is the trade, stated.

A recall measurement cannot justify a precision decision: adding -er moved the population figure by 0.21 links per query — invisible — and broke five negative controls. The controls are what caught it.